

Group Theory

Introduction

1.

- (a) The group of invertible $n \times n$ matrices over a finite field $\mathbb{F}_q$ is a finite group $\mathrm{GL}_n(\mathbb{F}_q)$. What is its order?
- (b) What about the special linear group $\mathrm{SL}_n(\mathbb{F}_q)$; what is its order?

2.

Let $1 \rightarrow A \rightarrow B \rightarrow C \rightarrow 1$ be a short exact sequence of groups. Show that if A and C are solvable, then B is also solvable.

3.

In this problem, fix an arbitrary finite nontrivial p -group G .

- (a) Show that $Z(G) \neq \{e\}$.
- (b) Show that G is nilpotent.

Rondo Capriccioso

* 4.

Let D_k be the dihedral group of order $2k$ where $k \geq 3$.

- (a) Show that the number of automorphisms of D_k is equal to $k \cdot \phi(k)$, where ϕ is Euler's ϕ -function.
- (b) Let $\mathrm{Aut}(D_k)$ denote the group of automorphisms of D_k . What is the structure of $\mathrm{Aut}(D_k)$? Describe the group as explicitly as you can.

* 5.

Let G be a group. By a maximal subgroup of G , we mean a subgroup $M \neq G$ such that the only subgroups containing M are M and G .

- (a) Describe all the maximal subgroups of the dihedral group of order $2p$ where p is an odd prime. How many are there?
- (b) Show that if a finite group G has only one maximal subgroup, then G is cyclic.
- (c) Show that if a maximal subgroup $M \triangleleft G$ is normal, then the index of M in G is finite and prime.

* 6.

If G is a group and $x, y \in G$, then we let $\langle x, y \rangle$ denote the subgroup of G generated by x and y . For $x \in G$, define $\text{Cyc}_G(x) := \{y \in G \mid \langle x, y \rangle \text{ is cyclic}\}$.

- (a) Show by example that $\text{Cyc}_G(x)$ need not be a subgroup of G .
- (b) Let $\text{Cyc}(G) = \bigcap_{x \in G} \text{Cyc}_G(x)$. Show that $\text{Cyc}(G)$ is a subgroup of G .
- (c) Show that $\text{Cyc}(G)$ lies in the center of G .
- (d) Assume now that G is finite. Show that $\text{Cyc}(G)$ is cyclic.